Machine Learning II

Data Science M.Sc.

Prof. Dr. Steffen Wagner
Berliner Hochschule für Technik

Winter term 2025/26

An Introduction to **Causal Machine Learning**

Definition

Causal Machine Learning is the intersection of Machine Learning and Causal Inference:

- Machine Learning focuses on predictive inference.
- Causal Inference focused on causal interpretation and mechanisms.

There are **two definitions of Causal ML**:

1. Using ML models to improve estimation of causal effects.
2. Using causal principles to improve ML models.

In the following, we will focus on the 1st definition and shortly discuss the 2nd perspective at the very end of the lecture.

Predictive Inference: SOTA

- **Predictive Inference:** using statistical or machine learning models to predict outcome Y based on observed patterns in training data $\mathbf{X}$.
- **Industry Standard** (for tabular data): **Gradient Boosted Trees** in combination with *Explainable AI (XAI)* (**SHAP**) to understand how predictions depend on observed feature values.

Example: User retention

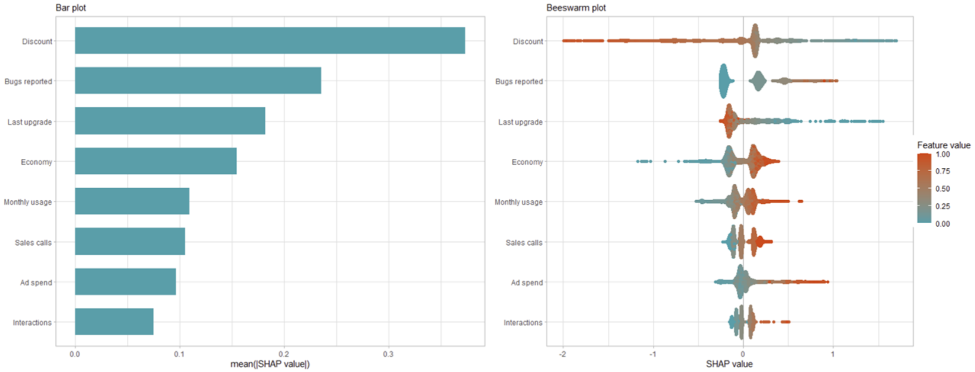
Business Case:

SaaS (Software as a Service) company wants to predict user retention based on the following data.

Variable	Description	Range_Units
Sales calls	Number of direct telephone calls made by the sales team to the client.	0 – 4 calls
Interactions	Total logged interactions (Calls + Emails + Meetings). Always >= Sales calls.	Count (Integer)
Economy	Macroeconomic index. Higher values indicate stronger economy.	0.0 (Recession) – 1.0 (Boom)
Last upgrade	Time elapsed since customer upgraded their software version.	Months (0 – 20)
Discount	The discount rate offered to the client for renewal.	0.0 (0%) – 0.5 (50%)
Monthly usage	Average monthly utilization rate of the software.	0.0 (None) – 1.0 (Max)
Ad spend	Budget allocated for user-specific marketing.	0.0 – 3.0 (Normalized Index)
Bugs reported	Number of software bugs/issues reported by the user.	Count (Integer)
Did renew	Target: Whether the client renewed contract (1=Yes, 0=No).	0 / 1

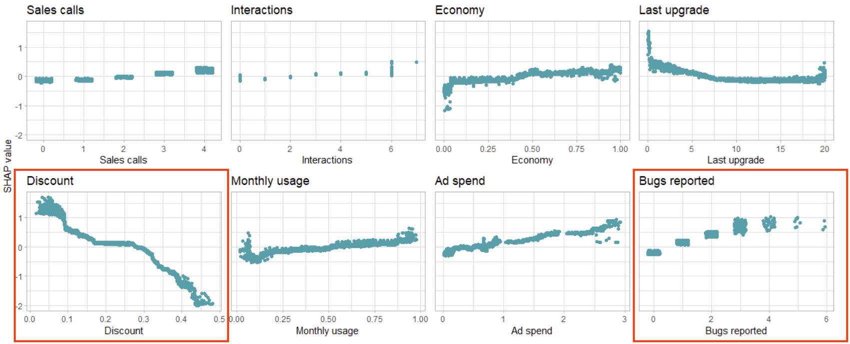
Example: User retention

Feature Importance



Example: User retention

Feature Dependence



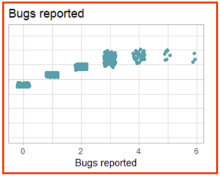
Findings:

- Users who report more bugs are more likely to renew?!
- Users with larger discounts are less likely to renew?!

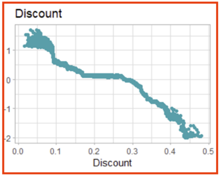
Example: User retention

Plausible Explanations:

- Users with high usage who value the product are more likely to report bugs and to renew their subscriptions.



- The sales force tends to give high discounts to customers they think are less likely to be interested in the product, and these customers have higher churn (i.e. lower renewal).



Conclusion: Perfect Setting for Predictive Inference!

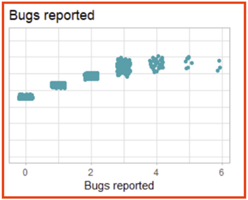
Example: User retention

Example: User retention

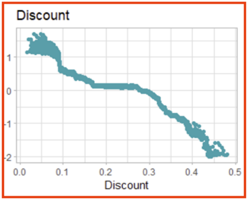
Example: User retention

Recommended Actions:

- Users who report more bugs are more likely to renew.
  $\Rightarrow$ **more bugs !!!**



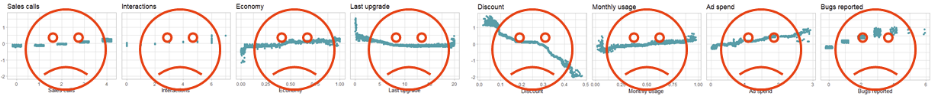
- Users with larger discounts are less likely to renew.
  $\Rightarrow$ **less discounts !!!**



Example: User retention

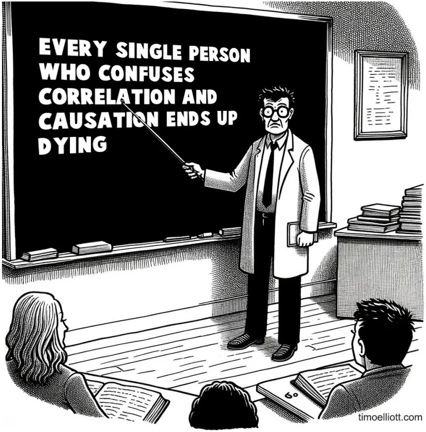
What happened?

- Your Predictive Task turned into a **Causal Task!**
- **Correlational Patterns** are not sufficient to reveal **Causal Relationships**!



Concepts for Causal Effect Estimation

Causation

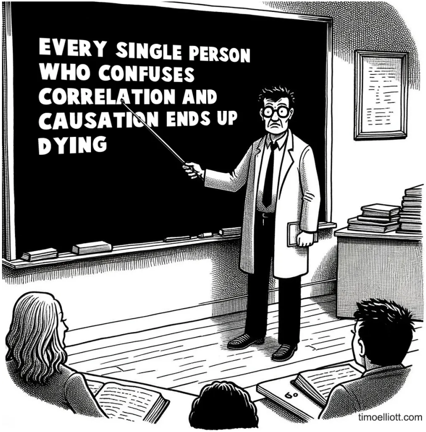

EVERY SINGLE PERSON
WHO CONFUSES
CORRELATION AND
CAUSATION ENDS UP
DYING
timoelliott.com

Causation

Minimum requirements for causation:

For X to cause Y

- X must precede Y
- X and Y must covary (vary together = correlation)
- No competing explanation can better explain the observed correlation.



Causal Analysis

Goal:

Inferential statistical statements about causal cause-and-effect relationships

Questions:

- What is the (causal) effect of X on Y?
- What is the expected difference in Y if X changes?

Traditional Tool: multiple linear regression

- **Causal effect** θ correspond to estimated regression coefficients.

Causal Analysis

Example:

How does athletic performance change with cigarette consumption?

Example: Push-ups Y vs. Cigarette Consumption X

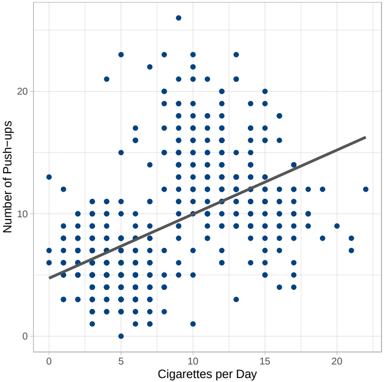
```
lm_Y_X <- lm(push_ups ~ cigarettes, data = df)
```

```
Call:
lm(formula = push_ups ~ cigarettes, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-9.6362 -2.8771 -0.5184  2.2171 16.5522

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.73584    0.38613   12.27   <2e-16 ***
cigarettes   0.52355    0.04089   12.80   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.344 on 498 degrees of freedom
Multiple R-squared:  0.2476,    Adjusted R-squared:  0.2461
F-statistic: 163.9 on 1 and 498 DF,  p-value: < 2.2e-16
```

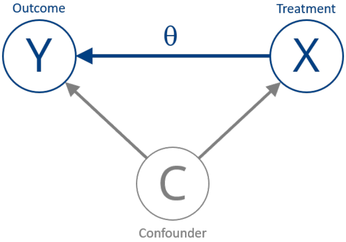


(Linear) model yields **implausible** results:

Cigarette consumption has a positive influence on athletic performance!

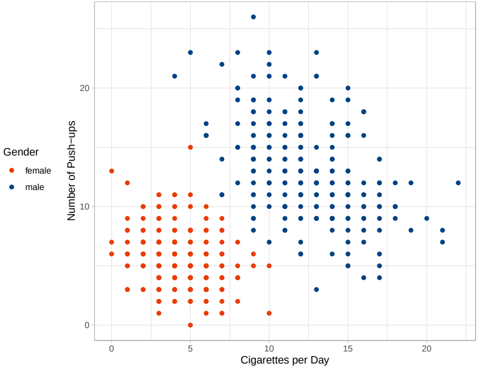
Confounder

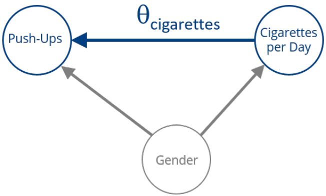
If there is a third influencing variable C that has a causal effect on X **and** Y, failing to account for it leads to a biased (= incorrect) effect estimate.



- Such a variable C is called a **Confounder**.
- Generally, multiple confounders $\boldsymbol{C} = \{C_1, C_2, \dots, C_i\}$ can occur, but this does not change the following discussion.

Example: Push-ups Y vs. Cigarette Consumption X





Confounding Variable/Confounder: **Gender**

- females smoke less $\Rightarrow$ smaller X values
- females do fewer push-ups $\Rightarrow$ smaller Y values

Confounder: Unbiased Effect Estimation

If we want to estimate the causal effect of X on Y **unbiasedly**, taking into account a confounder C (or a set of confounders), then C must be included in the modeling.

There are two strategies for this:

1. Inclusion of C in a joint model: $Y = f(X, C)$
2. Three-step procedure[1]:
   i) Model 1: Estimate the relationship: $C \rightarrow X$
   ii) Model 2: Estimate the relationship: $C \rightarrow Y$
   iii) Model 3: Effect estimation based on the residuals of the first two models.

$$\text{Resid}(C \rightarrow X) \rightarrow \text{Resid}(C \rightarrow Y)$$

[1] Frisch-Waugh-Lovell Theorem (1907 - 1963)

Accounting for Confounders

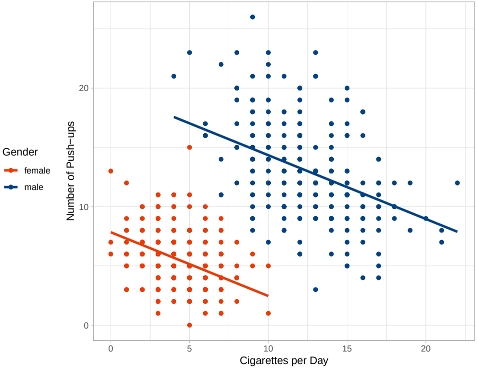
Strategy 1: Joint Model with `gender` as additional regressor

```
Call:
lm(formula = push_ups ~ cigarettes + gender, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-9.7234 -1.6961 -0.2251  1.4956 11.1241

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.83950    0.30088  26.055   <2e-16 ***
cigarettes  -0.53813    0.05422  -9.924   <2e-16 ***
gendermale  11.87957    0.51627  23.010   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.026 on 497 degrees of freedom
Multiple R-squared:  0.6357,    Adjusted R-squared:  0.6342
F-statistic: 433.7 on 2 and 497 DF,  p-value: < 2.2e-16
```

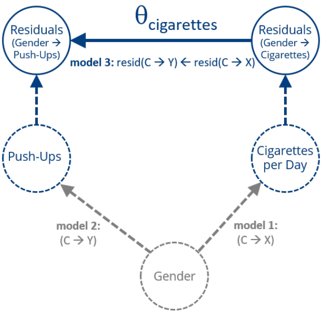


Results appear **plausible**:

Cigarette consumption has a negative influence on athletic performance.

Accounting for Confounders

Strategy 2: Multi-stage Estimation

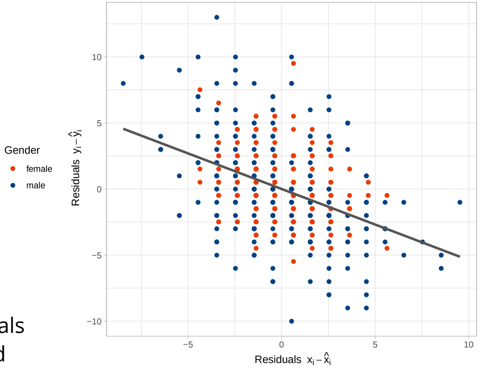


Accounting for Confounders

Strategy 2: Multi-stage Estimation

```
lm_X_C <- lm(cigarettes ~ gender, data = df)
lm_Y_C <- lm(push_ups ~ gender, data = df)
lm_resid <- lm(resid(lm_Y_C) ~ resid(lm_X_C))
```

```
                 Estimate  Std. Error t value Pr(>|t|)
(Intercept)    -1.7866e-16  1.3519e-01  0.0000        1
resid(lm_X_C) -5.3813e-01  5.4169e-02 -9.9343   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



- Result identical to joint estimation.
- The scatter plot clearly shows that the residuals from the two *preliminary* models are adjusted for the influence of the confounder.
- The residuals only contain the information whether someone (regardless of gender) achieves an above/below average number of push-ups or smokes an above/below average amount.

Spurious Correlation

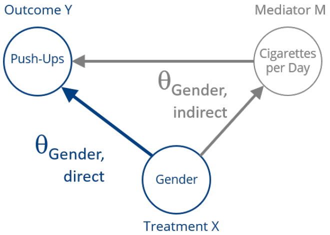
- In the causal analysis regarding the impact of cigarette consumption X we saw that the direction of the determined effect changed its sign:
  - without confounder C: positive relationship
  - with confounder C: negative relationship
- If the consideration of confounder variables C leads to a vanishing relationship ($\hat{\theta} = 0$) between X and Y, the observed *unconfounded* relationship is called a **Spurious Correlation**.
- Spurious correlations occur pretty easily in time series data, where a common temporal trend (= confounder C) is responsible for spurious correlations[2].

[2] many examples: `https://www.tylervigen.com/spurious-correlations`

Mediators

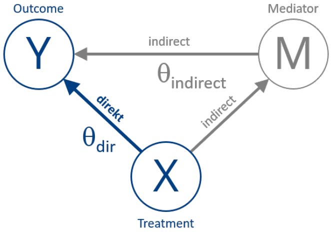
Alternative Question:

How large is the influence of gender on the number of push-ups?



- We see that gender has a **direct** and **indirect** (via the causal effect on cigarette consumption) influence on the target variable.
- In this question, cigarette consumption is not a *confounder* but a **mediator**.

Mediators: Unbiased Effect Estimation

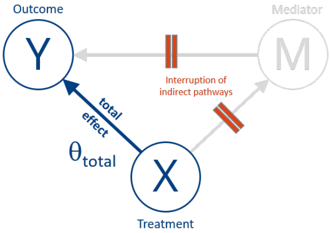


To estimate the influence of a treatment X on a target variable Y in the presence of a mediator M unbiasedly, there are again two possible approaches:

1. Direct estimation of the total effect $\hat{\theta}_{tot}$
2. Path analysis $\hat{\theta}_{tot} = \hat{\theta}_{dir} + \hat{\theta}_{ind}$

Approach 1

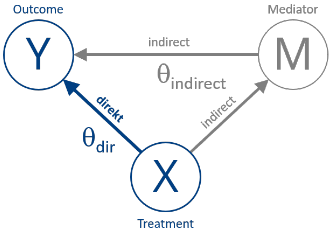
Direct estimation of the total effect



- Important: **no** consideration of M in the regression model
- Estimation of a **bivariate** model according to $Y = \beta_0 + \theta_{tot} \cdot X$

Approach 2

Path Analysis



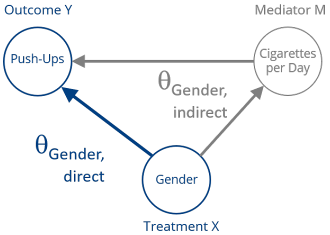
i. Estimation of the direct effect $\hat{\theta}_{dir}$ and the effect of M on Y via multiple regression $Y = \beta_0 + \theta_{dir} \cdot X + \theta_{M \to Y} \cdot M$

ii. Bivariate estimation of the effect of X on M via $M = \beta_0 + \theta_{X \to M} \cdot X$

iii. Determination of the total effect[3] $\hat{\theta}_{tot} = \hat{\theta}_{dir} + \underbrace{\hat{\theta}_{X \to M} \cdot \hat{\theta}_{M \to Y}}_{\hat{\theta}_{indirect}}$

[3] **Important:** the relationship given here only applies to linear dependencies and not generally.

Example: Push-ups Y vs. Gender X



To estimate the influence of gender X on the number of observed *push-ups* Y in the presence of the mediator *cigarette consumption* M unbiasedly, we consider both approaches:

1. Direct estimation of the gender effect $\hat{\theta}_{tot}$
2. Path analysis $\hat{\theta}_{tot} = \hat{\theta}_{dir} + \hat{\theta}_{ind}$

Example: Push-ups Y vs. Gender X

Approach 1: **Direct Estimation** of the Total Effect

```r
lm_Y_X <- lm(push_ups ~ gender, data = df)
summary(lm_Y_X)
```

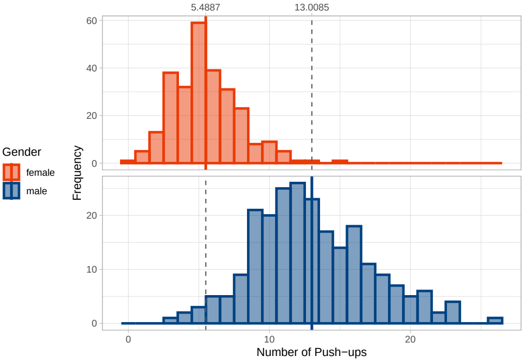
```
Call:
lm(formula = push_ups ~ gender, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-10.0085  -2.0085  -0.4887   1.6313  12.9915

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.4887     0.2029   27.05   <2e-16 ***
gendermale    7.5198     0.2966   25.36   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.309 on 498 degrees of freedom
Multiple R-squared:  0.5635,    Adjusted R-squared:  0.5626
F-statistic:   643 on 1 and 498 DF,  p-value: < 2.2e-16
```

Total Effect: $\hat{\beta}_{total} = \overline{Y}_{male} - \overline{Y}_{female} = 13.0085 - 5.4887 = 7.5198$



Example: Push-ups Y vs. Gender X

Approach 2: **Path Analysis**

```
lm_Y_XM <- lm(push_ups ~ gender + cigarettes, data = df)
```

```
             Estimate Std. Error t value  Pr(>|t|)
(Intercept)  7.839498   0.300881 26.0552 < 2.2e-16 ***
gendermale  11.879570   0.516269 23.0104 < 2.2e-16 ***
cigarettes  -0.538129   0.054223 -9.9243 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- $\hat{\theta}_{dir}$: for the same cigarette consumption, males achieve on average 11.88 more push-ups.
- $\hat{\theta}_{M\to Y}$: per additional cigarette smoked, one achieves 0.54 fewer push-ups (compare slide 21)

```
lm_M_X <- lm(cigarettes ~ gender, data = df)
```

```
            Estimate Std. Error t value  Pr(>|t|)
(Intercept)  4.36842    0.15333  28.491 < 2.2e-16 ***
gendermale   8.10166    0.22413  36.148 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- $\hat{\theta}_{X\to M}$: Males smoke on average 8.1017 cigarettes more per day.

Total Effect: $\hat{\theta}_{tot} = \hat{\theta}_{dir} + \underbrace{\hat{\theta}_{X\to M} \cdot \hat{\theta}_{M\to Y}}_{\hat{\theta}_{indirect}} = 11.8796 + 8.1017 \cdot (-0.5381) = 7.5198252$

Basics of Causal Effect Estimation

Summary:

- Without knowledge of the causal relationships, there is a danger that the estimated effects are biased and thus incorrect.
- Depending on the causal relationship between the considered variables (*Treatment*, *Confounder*, *Mediator* or *Collider*[4]), the analytical approach differs.
- Linear regression models represent an important set of methods for conducting causal analyses, i.e., for estimating causal effects.

But: the assumption of linear relationships is very strong!

[4] not covered: check `https://en.wikipedia.org/wiki/Collider_(statistics)`

Machine Learning models for Causal Effect Estimation

Machine Learning models for Causal Effect Estimation

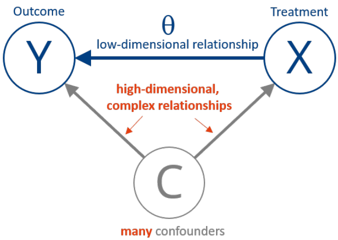
Why Machine Learning models?

- very flexible and powerful functions (interactions, non-linearities)
- no problems dealing with lots of features (= wide data)
- (almost) no assumptions on functional relationship:
  $\Rightarrow$ avoids model misspecification

Machine Learning models for Causal Effect Estimation

Goal:

- We want to estimate a low-dimensional causal effect θ in the presence of high-dimensional and complex confounding relationships $Y(C)$ and $X(C)$.



- We want to model the confounding relationships $Y(C)$ and $X(C)$ with modern ML models.

Machine Learning models for Causal Effect Estimation

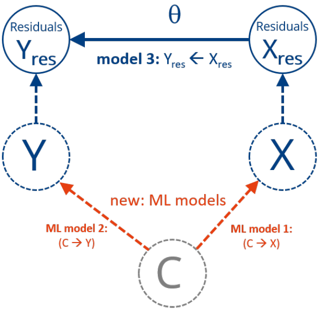
But: ML models have been developed for predictive tasks!

- Since ML models tend to overfit the data, regularization is required.
- Regularization reduces overfitting but leads to a **regularization bias** for estimated effects and relationships.
- The model *shrinks* the effect of confounding variables C too much.
- In other words: the **ML model fails to fully capture the confounding effect** of C with respect to both treatment X and outcome Y.
- This means the estimated causal effect θ is still biased because of the non-captured, remaining confounding.

Therefore the joint modelling approach $Y = f(X, C)$ based on a *single* ML model f considering *confounders* C and treatment X cannot be applied because it would lead to a biased causal effect estimation.

The Double ML approach[5]

- **Double ML** provides a scheme for an unbiased estimation of θ using ML algorithm to model the confounding relationships.
- It is shown that the **regularization bias vanishes**, when the causal effect is estimated on the residuals Y_{res} and X_{res}.
- In addition: an **overfitting bias** in model 3 has to be dealt with.
- The proofs are rather mathematical than intuitive.

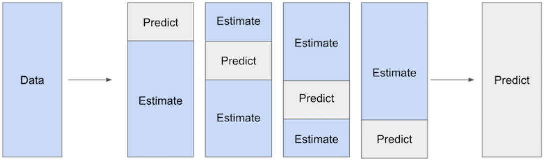


To get an **unbiased causal effect estimate** a so-called **Cross-Fitting** approach has to be applied.

[5] Chernozukov et al. (2018): "Double/Debiased Machine Learning for Treatment and Structural Parameters." The Econometrics Journal, 21(1), C1–C68. `https://doi.org/10.1111/ectj.12097`

The Double ML algorithm

1. Split the data into K folds
2. Train two ML models on $K-1$ folds:
   a) model 1 $C \rightarrow X$: outcome X, features $\boldsymbol{C}$
   b) model 2: $C \rightarrow Y$: outcome Y, features $\boldsymbol{C}$
3. Use the models to make *predictions* ($\hat{X}$ and $\hat{Y}$) on the *held-out fold*: **Cross-Fitting**



4. Compute residuals as $\hat{X}_{res} = X - \hat{X}$ and $\hat{Y}_{res} = Y - \hat{Y}$
5. Use a third regression model to estimate causal effect from residuals:
   Regress $\hat{Y}_{res}$ on $\hat{X}_{res}$, obtain the coefficient $\hat{\theta}$
6. Repeat for all folds, average resulting coefficients to obtain the final causal estimate.

Does the choice of the ML algorithm matter?

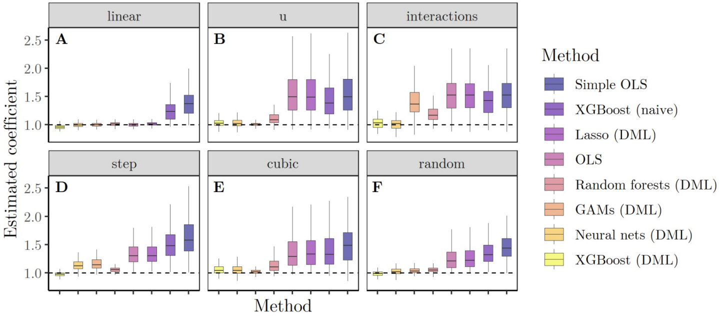


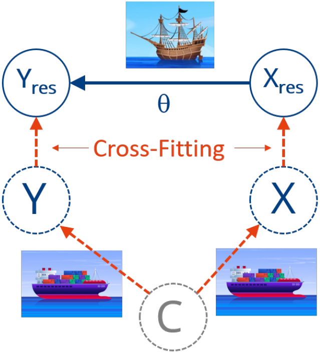
Figure 5: Results for Case 1 - distribution of estimated coefficients for each method across 100 simulations by functional form (outliers not displayed). The dashed line marks the true causal effect ($\beta = 1$). **A** Linear confounding. **B** U-shaped/squared confounding. **C** Pairwise interactions between confounders. **D** Confounding via step function. **E** Cubic confounding. **F** Confounding functional form drawn randomly for each confounder.

taken from: Fuhr, J., Berens, P., & Papies, D. (2024). Estimating Causal Effects with Double Machine Learning–A Method Evaluation. `https://arxiv.org/pdf/2403.14385`.

The Double ML approach

Summary

- Double ML provides a framework for estimating causal effects using **ML algorithms** modeling complex nuisance relationships.
- **Statistical model** estimates causal effect based on residuals.
  - statistical inference !!!
  - confidence intervals for estimates !!!



Example: User retention

Example: User retention

Example: User retention

The Data

SaaS (Software as a Service) company wants to predict user retention based on the following data.

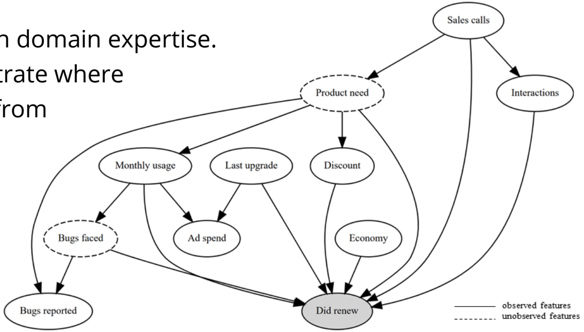
Variable	Description	Range_Units
Sales calls	Number of direct telephone calls made by the sales team to the client.	0 – 4 calls
Interactions	Total logged interactions (Calls + Emails + Meetings). Always >= Sales calls.	Count (Integer)
Economy	Macroeconomic index. Higher values indicate stronger economy.	0.0 (Recession) – 1.0 (Boom)
Last upgrade	Time elapsed since customer upgraded their software version.	Months (0 – 20)
Discount	The discount rate offered to the client for renewal.	0.0 (0%) – 0.5 (50%)
Monthly usage	Average monthly utilization rate of the software.	0.0 (None) – 1.0 (Max)
Ad spend	Budget allocated for user-specific marketing.	0.0 – 3.0 (Normalized Index)
Bugs reported	Number of software bugs/issues reported by the user.	Count (Integer)
Did renew	Target: Whether the client renewed contract (1=Yes, 0=No).	0 / 1

Example: User retention

The data generating process is reflected in a **Causal Graph:**

Best Practice:

- Create a causal graph based on domain expertise.
- The causal graph helps to illustrate where predictive relationships differ from causal relationships.

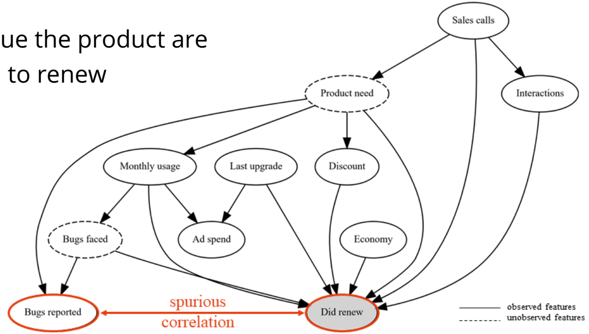


Example: User retention

A Causal Graphs helps to understand predictive relationships.

Interpretation:

- Users with high usage who value the product are more likely to report bugs and to renew their subscriptions.
- **Spurious Correlation** due to unobserved confounder!

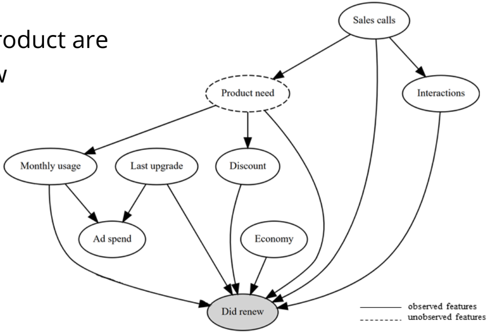


Example: User retention

A Causal Graphs helps to understand predictive relationships.

Interpretation:

- Users with high usage who value the product are more likely to report bugs and to renew their subscriptions.
- **Spurious Correlation** due to unobserved confounder!
- **Removal** from causal model!

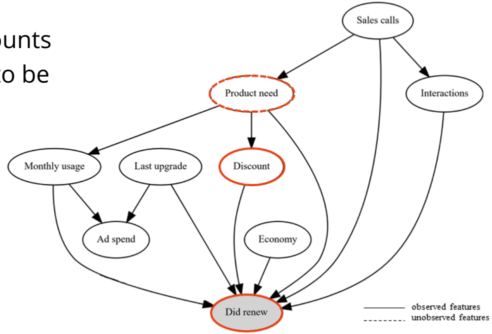


Example: User retention

A Causal Graphs helps to understand predictive relationships.

Interpretation:

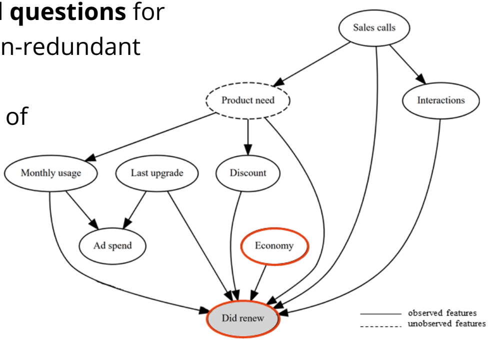
- The sales force tends to give high discounts to customers they think are less likely to be interested in the product, and these customers have higher churn.
- **Biased Effect** due to *unobserved* confounder!



Example: User retention

Identification of independent components

- **Predictive models can answer causal questions** for features that carry independent, i.e. non-redundant information $\Rightarrow$ no confounder bias.
- Data-driven approach for identification of independent features: analysis of **redundancy structure**

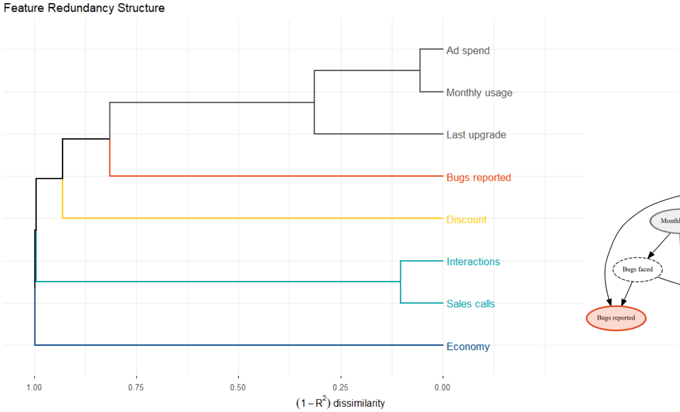


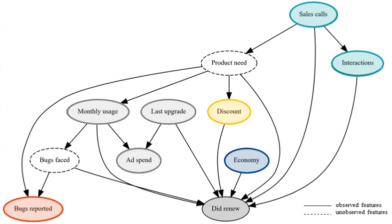
Feature Redundancy

Idea:

- Redundant features explain in a **univariate model** the outcome Y in a similar way.
- The **squared correlations $\mathbf{R^2_{jk}}$** between predictions $\hat{Y}_j = f(X_j)$ and $\hat{Y}_k = f(X_k)$ from univariate models for features X_j and X_k serve as similarity measure.
- Low R^2_{jk} means low similarity means low redundancy.
- The pairwise similarity R^2_{jk} between two features serves as similarity matrix for **Hierarchical Feature Clustering**.
- **Isolated features**, i.e. clusters, are identified to be **independent**.

Feature Redundancy



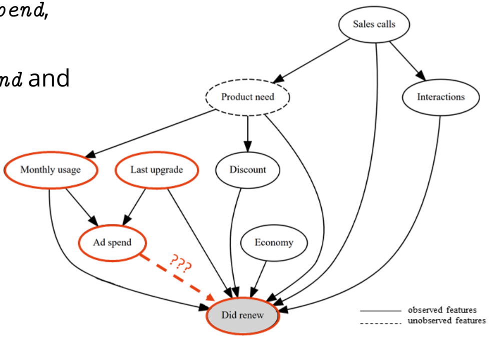


Example: User retention

Analysis of feature redundancy structure

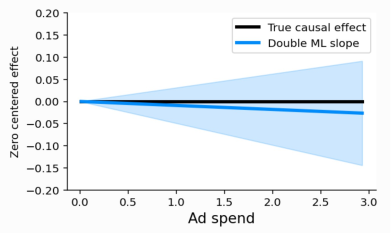
- High similarity observed between `Ad spend`, `Monthly usage` and `Last update`.
- Observed relationship between `Ad spend` and outcome `Did renew` might be due to confounding.

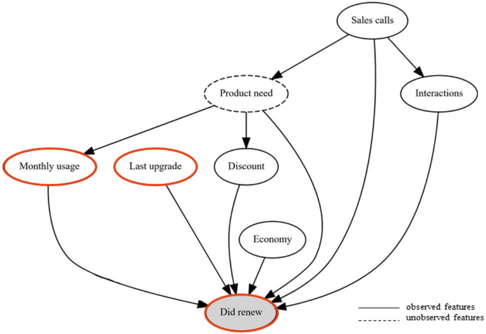
Analysis with **Double ML** approach!



Example: User retention

Causal Analysis of $\mathit{Ad\ Spend}$ with Double ML:





No causal effect!

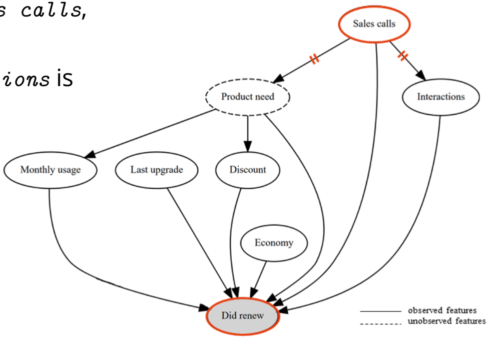
$\Rightarrow$ **Removal** from Causal Model!

Example: User retention

Analysis of feature redundancy structure

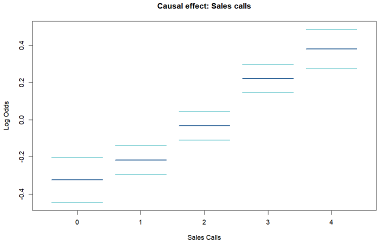
- High similarity observed between `Sales calls`, and `Interactions`.
- Based on the Causal Diagram `Interactions` is a *Mediator* for the `Sales calls` feature.

To determine the total `Sales calls` effect, a *univariate* model should be estimated!

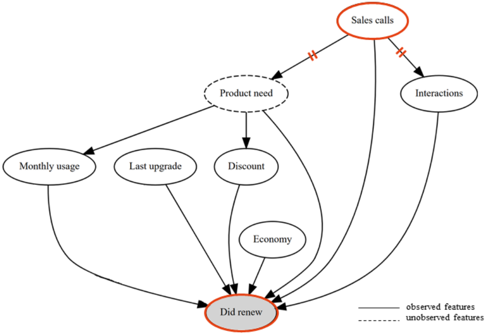


Example: User retention

Univariate analysis of feature `Sales calls`



$\Rightarrow$ **Significant causal effect** determined!



Example: User retention

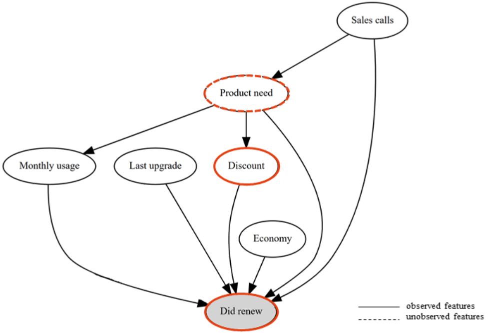
Example: User retention

Analysis of feature redundancy structure

Unobserved confounders:

- Strongly biased estimates due to unobserved confounder!
- Unobserved confounder means: **You don't have the data!**

(Randomized) experiments required to get data!
(= gold standard of causal analysis)



Summary

Machine Learning models for causal analysis:

- Predictive models based on correlations are fine for Predictive tasks.
- Causal tasks require causal models.
- Confounding (=nuisance) variables can cause spurious correlations and have to be identified based on domain expertise (= causal graphs).
- The Double Machine Learning approach combines traditional concepts of causal analysis with modern ML algorithms.
- Make ese of **Doube ML** to correct for observed confounding and eliminate spurious correlations.
- **But:** confounders have to be observed / measured / available.
- Unobserved confounders require randomized experiments.

Software packages & Literature

Further reading:

- Scott Lundgreen (2021): Be careful when interpreting predictive models in search of causal insights

  https://shap.readthedocs.io/en/latest/example_notebooks/overviews/Be%20careful%20when%20interpreting%20predictive%20models%20in%20search%20of%20causal%20insights.html
- Matheus Facure Alves (2022): Causal Inference for The Brave and True

  https://matheusfacure.github.io/python-causality-handbook/landing-page.html

Software:

- DoubleML (Python and R): `https://docs.doubleml.org/`
- EconMl (only Python): `https://econml.azurewebsites.net/index.html`
- CausalML (only Python): `https://causalml.readthedocs.io/`